Assignment2 – Bayesian Networks

Given: April 29 Due: May 4

Problem 2.1 (Probabilities in Python)

Complete the partial implementation of probabilities at `https://kwarc.info/teaching/AI/resources/AI2/probabilities/`

Solution: See `https://kwarc.info/teaching/AI/resources/AI2/probabilities/`

Problem 2.2 (Bayesian Rules)

Give the formulas and a one-sentence explanation of the following basic rules in Bayesian inference:

1. Bayes rule

Solution: $P(A \mid B) = P(B \mid A)P(A)/P(B)$
The conditional probability of A given B multiplied by the probability of B is the same as the conditional probability of B given A multiplied by the probability of A – both are equal to $P(A,B)$. We can use that to compute one conditional probability from the other.

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2. Product rule

Solution: $P(A,B) = P(A \mid B)P(B)$, The probability of A and B is the product of the probability of B and the one of A given B. If A and B are independent, this simplifies to $P(A,B) = P(A)P(B)$.

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3. Chain rule

Solution: $P(A_1, \ldots, A_n) = P(A_n \mid A_{n-1}, \ldots, A_1) \cdot P(A_{n-1} \mid A_{n-2}, \ldots, A_1) \cdot \ldots$
Iterated application of the product rule.

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4. Marginalization

Solution: Marginalization of A with respect to Y: $P(A) = \sum_{y \in E} P(A, y)$ where E is the set of values of Y. Since the probabilities of the values of Y sum to 1, we can always introduce/remove a sum over all values.

$P(X, e) = \langle P(x_1, e), P(x_2, e), P(x_3, e) \rangle = \langle 0.2, 0.3 \rangle$

$= 0.5$

$\alpha = \frac{1}{\sum_i P(x_i, e)} = \frac{1}{0.5} = 2$

$\times 2$

$\langle 0.4, 0.6 \rangle$

5. Normalization

Solution: Normalization of X with respect to event e: $P(X \mid e) = \alpha(P(X, e))$ where α is the function that multiplies every element in a vector v (here: the vector $\langle P(X = x_1, e), \ldots, P(X = x_n, e) \rangle$ where the x_i are the possible values of X) by $1/\sum_i v_i$. The probability of X given e can be obtained by normalizing the joint probability X and e.

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m morning
o in office
¬m evening/night
¬o not in office

Problem 2.3 (Is your TA in the office?)

You want to discuss something with your TA. You know that

1. the probability of your TA being in the office, assuming it is morning, is $\frac{1}{5}$,
2. if your TA is in the office, there is a $\frac{1}{3}$ probability it is morning, (Assumption)
3. the probabilities that it is morning or afternoon are both $\frac{1}{2}$

Your tasks:

1. Write down the probabilities mentioned above as formulas

Solution: Let m denote that it is morning and o denote that the TA is in the office.

1. $P(o \mid m) = \frac{1}{5}$
2. $P(m \mid o) = \frac{1}{3}$
3. $P(m) = P(\neg m) = \frac{1}{2} = \frac{5}{10}$

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	m	¬m	
o	1/10	2/10	3/10
¬o	4/10	3/10	7/10
	5/10	5/10 = 10/10	

$P(o, m) = P(o \mid m) \cdot P(m)$

$1/5 \cdot 1/2 = 1/10$

$P(o, m) = P(m) - P(\neg o, m) = 5/10 - 1/10 = 2/10$

$P(o, \neg m) = P(o) - P(o, m)$

$P(\neg o, \neg m) = P(\neg m) - P(\neg o, m)$

$\frac{5}{10} - \frac{2}{10} = \frac{3}{10}$

2. Compute the full joint probability distribution

Solution:

- $P(o, m) = P(o \mid m) \cdot P(m) = \frac{1}{10}$ *(product rule)*
- $P(\neg o, m) = \frac{4}{10}$, because $P(m) = P(o, m) + P(\neg o, m)$ *(marginalization)*

Now, from $P(m \mid o) \cdot P(o) = P(m, o)$ it follows that $P(o) = \frac{\frac{1}{10}}{\frac{1}{3}} = \frac{3}{10}$. So we get

- $P(o, \neg m) = \frac{2}{10}$, because $P(o) = P(o, m) + P(o, \neg m)$ *(marginalization)*
- $P(\neg o, \neg m) = \frac{3}{10}$, because $1 - P(o) = P(\neg o) = P(\neg o, m) + P(\neg o, \neg m)$ *(marginalization)*

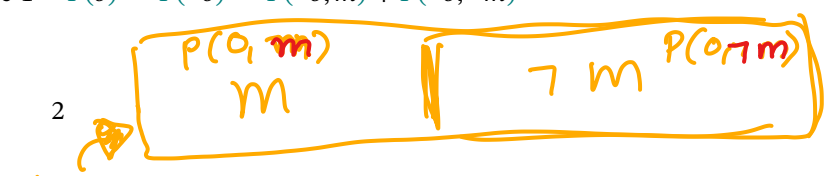
$P(m \mid o) = 1/3$

$P(m \mid o) = \frac{P(m, o)}{P(o)}$

$\frac{1}{3} = \frac{1/10}{P(o)}$

$P(o) = \frac{3}{10}$

$P(o, m) = 1/10$



3. What's the probability you'll meet your TA, if you come to the office in the afternoon?

Solution: $P(o \mid \neg m)\}c = \frac{P(o,\neg m)}{P(\neg m)} = \frac{4}{10}$

Problem 2.4 (AFT Tests)

Trisomy 21 (*Down syndrome*) is a genetic anomaly that can be diagnosed during pregnancy using an amniotic fluid test.

The probability of a foetus having Down syndrome is strongly correlated with the age of the pregnant parent. We will only consider the following two age groups.

1. For 25 year olds the probability is one in 1250,
2. for 43 year old parents it increases to one in fifty.

However, diagnostic tests are never perfect. We distinguish two kinds of errors:

3. Type I Error (False Positive): The test result is positive even though the child is healthy.
4. Type II Error (False Negative): The test result is negative even though the child has trisomy 21.

The probabilities of Type I and Type II Errors are both merely 1% for amniotic fluid tests for Down syndrome.

1. Express the four items above in the form of conditional probabilities. Use the random variable F with domain $\{Age_{25}, Age_{43}\}$ for the age of the pregnant person and the Boolean random variables *Pos* and *Down* for the propositions "*The amniotic fluid test is positive*" and "*The child has Down syndrome*" respectively.

Solution: $P(Down \mid F = Age_{25}) = 0.0008$, $P(Down \mid F = Age_{43}) = 0.02$, $P(Pos \mid \neg Down) = 0.01$, $P(\neg Pos \mid Down) = 0.01$.

2. Assume that we have a 25 year old pregnant person. Using Bayes' theorem, express and compute the probability that their child has Down syndrome, given that the amniotic fluid test is positive. What can we conclude from the result?

Solution: We fix $F = Age_{25}$, making $P(Down) = 0.0008$ and compute:

(same calculation below)

Example lets say we have

100,000 ~ 25yrolds

× 0.0008 = 80 actual expectation

without 99920 healhy

False Positive: 999.2 (0.01)

Positive = 79.2+ = 1078.4 postivitive test

$$P(Down \mid Pos) = \frac{P(Pos \mid Down) \cdot P(Down)}{P(Pos)} = \frac{P(Pos \mid Down) \cdot P(Down)}{P(Pos \wedge Down) + P(Pos \wedge \neg Down)}$$
$$= \frac{P(Pos \mid Down) \cdot P(Down)}{P(Pos \mid Down) \cdot P(Down) + P(Pos \mid \neg Down) \cdot P(\neg Down)}$$
$$= \frac{(1 - P(\neg Pos \mid Down)) \cdot P(Down)}{(1 - P(\neg Pos \mid Down) \cdot P(Down)) + P(Pos \mid \neg Down) \cdot (1 - P(Down))}$$
$$= \frac{0.99 \cdot 0.0008}{0.99 \cdot 0.0008 + 0.010.9992} \approx 0.07$$

So, even with a positive test result, the probability of the child actually having Down syndrome is still only 7%, simply due to Down syndrome being relatively rare in young parents. Consequently, there is little point in applying this particular test without exceptional cause for concern.

Problem 2.5 (Bayesian Networks)

Consider the following situation about an alarm clock:

- It fails to ring if its batteries are empty.
- It fails to ring if it the volume is muted.
- You might oversleep if your alarm clock fails.
- You might oversleep if you stay up late.

You want to model this situation as a Bayesian network using Boolean random variables.

E → F ← M
F → O ← U

1. Give an appropriate set of random variables and their meaning and draw the Bayesian network (using an appropriate variable order).

Solution: Variables: F (clock fails), E (batteries empty), M (clock on mute), O (oversleep), U (up late).
Network: $E \to F \leftarrow M$ and $F \to O \leftarrow U$

$P(E, M, F, U, O) =$
$P(E)\, P(M)\, P(F \mid E, M)\, P(U)\, P(O \mid F, U)$

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2. Give the probability of the clock failing in terms of the entries of the conditional probability tables of your network.

Solution: $P(F) = \sum_{e,m \in \{true, false\}} P(F \mid E = e, M = m) \cdot P(M = m) \cdot P(E = e)$

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e → empty battery
↳ true/false

m → muted {t,f}

Now you decide to model the failing of the clock as a deterministic node.

3. Explain (in about 3 sentences) whether that decision is justified by the description, and how it changes the conditional probability tables of your model.

$P(F \mid E, M)\, P(E)\, P(M)$
$+ P(F \mid E, \neg M)\, P(E)\, P(\neg M)$
$+ P(F \mid \neg E, M)\, P(\neg E)\, P(M)$
$+ P(F \mid \neg E, \neg M)\, P(\neg E)\, P(\neg M)$

E
f
f
t
t

M
f
t
f
t

Failure to Ring
f
t
t
t

Solution: In that case, $F = M \vee E$, and we do not have to store a conditional probability table for F and only need to store that definition.
It is not justified. The description does imply that F holds if M or E does. But the description does not exclude that F holds even if neither M not E does (e.g., if the alarm clock is defective).

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4. State the probability of the clock failing in terms of the entries of the conditional probability tables of your network (with the deterministic node for the failing clock).

Solution: $P(F) = 1 - P(\neg M) \cdot P(\neg E)$

alarm does not fail if $\neg E \wedge \neg M$

$P(F) = 1 - P(\neg F)$

$\downarrow$

$1 - P(\neg E, \neg M)$

$1 - P(\neg E) \cdot P(\neg M)$